

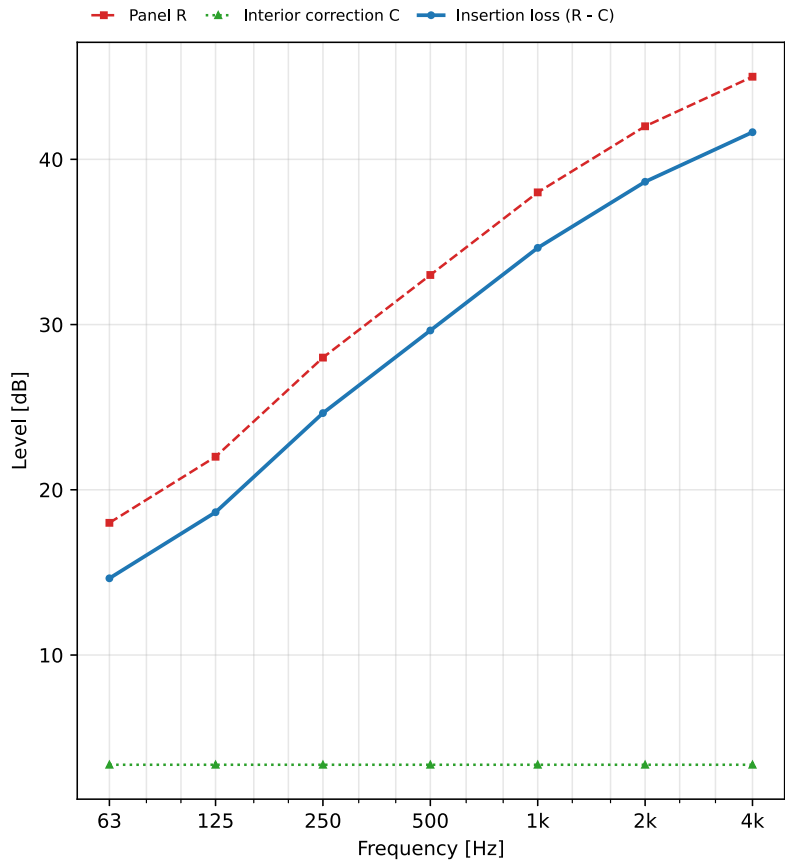
Machine enclosure insertion loss

Predicted insertion loss of a machine enclosure, estimated from the panel transmission loss and the interior build-up correction (Bies, Hansen & Howard, Engineering Noise Control 5th ed., section 7.4.2, Eqs. (7.103) and (7.111)). This is a prediction from the declared data, not a measurement of the enclosed machine.

Noise source:	Sheet-steel close-fitting machine enclosure (design case)	Test environment:	Machine hall, line 3 (design case)
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Octave-band insertion loss

f [Hz]	R [dB]	C [dB]	IL [dB]
63	18.0	3.4	14.6
125	22.0	3.4	18.6
250	28.0	3.4	24.6
500	33.0	3.4	29.6
1000	38.0	3.4	34.6
2000	42.0	3.4	38.6
4000	45.0	3.4	41.6



Mean insertion loss IL = **28.9 dB**

Mean panel transmission loss R = 32.3 dB
External surface area $S_E = 24.00 \text{ m}^2$
Internal surface area $S_i = 30.00 \text{ m}^2$

Predicted (estimated) result computed from the declared enclosure geometry, its panel transmission loss and the interior absorption; it is not a measurement. The realised insertion loss also depends on leaks, structural flanking and machine-to-enclosure contact, which this model does not represent.

Result vs requirement: IL = 28.9 dB, required $\geq 20.0 \text{ dB}$ → **PASS**

$IL = R - C$ with the interior build-up correction $C = 10 \lg(0.3 + S_E / R_i)$ and the interior room constant $R_i = S_i \alpha_i / (1 - \alpha_i)$ (Eqs. (7.103) and (7.111)). The panel transmission loss R is supplied by the caller (measured or predicted); this evaluation combines it with the interior build-up correction only.

■ The results are a prediction computed from the stated inputs; they are not a measurement of a specimen.
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